Hockey Analytics League Operations (HALO) Hackathon
Presented to Sportlogiq & Teamworks

Quantifying the Impact of Offensive Off-Puck Positioning Using Pass Networks

By

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Introduction

In professional sports, evaluating player performance is a complex and time-sensitive task. Coaches and analysts must assess individual contributions and overall impact under significant time constraints, often relying on limited observations or traditional statistics. This is why over the past decades, technological advancements have transformed this evaluation process. Player-tracking systems (Kovalchik, 2023) now allow analysts to process vast amounts of game data, while structural and network-based analytical methods (Korte, 2018) (Korte, 2019) provide deeper tactical insights into team dynamics.

Despite the increasing availability of advanced performance metrics, certain players continue to be undervalued. Those who do not rely on flashy plays often have an impact that is subtle but structurally important. These contributions are often indirect and may not immediately result in measurable outcomes such as goals. Instead, they manifest through rapid puck progression, increased spatial occupation, and increased passing availability, which are aspects that require structural analysis to be properly quantified.

One method that particularly piqued our interest is passing network analysis, which has been increasingly applied in soccer to better understand the interdependent relationships between players within a team. (Korte, 2019) From these networks, various centrality metrics can be computed to quantify and compare players' influence on the quality of their team's puck flow as well as their overall involvement in play progression. (Lamberts, 2024) (McHale, 2018) Another metric widely used in other sports is expected goals (**xG**) models that estimate the scoring probability of a shot, which have provided a probabilistic foundation for evaluating offensive actions. Extending this idea, passes can be assessed through their risk–reward trade-off, motivating possession added value (**PAV**) frameworks that quantify how individual passes shift scoring probability. (Tyrelstokes, 2021) (Tyrelstokes, 2022) To further assess the quality of a pass or positioning, we can use a pass risk prediction model to the likelihood of the pass to be completed (Felices, 2023).

Beyond visualizing team structure, passing networks allow centrality metrics to be extracted to quantify player roles within the collective system. They allow us to measure how frequently a player is involved in puck circulation, how important they are to possession flow, which can give important insight in players whose impact may not appear in traditional statistics such as goals or assists. (Lamberts, 2024) (McHale, 2018) **Passing centrality** reflects how frequently a player is involved in passing sequences, indicating their direct participation in team play. (Lamberts, 2024) **Eigenvector** centrality captures how well a player is connected to other influential and valuable players in the team. (Lamberts, 2024) **Betweenness** centrality measures how often a player acts as an intermediary in the flow of possession, which reflects how crucial they are in linking different players during play. (Lambert, 2024) **PageRank** centrality quantifies a player's influence in the network by considering both the frequency and the importance of passes they receive, rewarding players who are integral to key passing chains. (López Peña, 2012)

Inspired by these findings, we hypothesized that a similar network-based analysis could provide insight into off-the-puck contributions in hockey. Additionally, to capture the relative importance of each pass, we decided to incorporate a measure of its potential impact on scoring, similar to expected goals in shot analysis. This allows each pass to carry a value, reflecting how much it contributes to creating scoring opportunities. (McHale, 2018) By combining network centrality measures with these weighted pass values, we define an **Off-Puck Impact Score** (**OPIS**) to quantify the indirect contributions of players who facilitate play, maintain possession, and enable teammates, which are, in our opinion, aspects that are often overlooked in traditional statistics. Moreover, weighting edges by possession value added

(PAV) allows the network to capture not only the frequency of player interactions but also the offensive impact of each pass.

Methods and results

The data we used was event and tracking data from Sportlogiq during the 2023-2024 AHL regular season. To simplify the pass structures and analysis of network metrics, we decided to split the data keeping only passes that start and end in the offensive zone. Hockey teams' passing structure is much clearer and well executed when looking at this section of the data (Pitassi, 2025).

The first step in our analysis is to compute value weights for each pass. To achieve this we decided to associate a **Possession Added Value (PAV)** to each pass. For simplicity and because of time constraints, this was done by assuming that the only alternative to the pass would be to shoot it (compute Xg at the start of the pass) and that there could be a shot directly at the time of the reception (compute Xg at the end of the pass). To determine **PAV**, we took the difference between these two values (Xg_after - Xg_before) (Tyrelstokes, 2021)(Tyrelstokes, 2022).

In order to build our **PAV** weights pass we needed to build an expected goal model **(Xg)**. We decided to use an Extreme Gradient Boosting (XgBoost) machine learning model. The model was trained on 5 features: distance, angle, nearest defender distance, number of defenders in the lane, game state (powerplay or not) and the pass before if the shot is within 2 seconds of the pass. The distance feature is the total distance from the point that the shot was taken to the goal (coordinates 89,0) (Younggren & Younggren, 2021). The shot angle is calculated relative to the center of the goal. The nearest defender was computed by using the tracking data of all defenders and taking the minimum distance from the shooter. The number of defenders in the lane was found by taking the line from the shooter to the net and taking the distance from every defender from that line. We only kept shooters within 2 feet of the shooting lane. To determine if the shot was taken on powerplay we checked if the shooting team had more skaters than the defending team. Finally to quantify the impact of the pass within 2 seconds before the shot we used a combination of K-nearest-neighbor (KNN) and K-means clustering. We sampled all of the passes in the offensive zone ($x > 25$) and used sklearn.cluster's KMeans library to find 20 clusters differentiating the passes. The features of the cluster are: x, y, cos, sin and length. We allowed the program to run 20 times to find the best fit clustering possible. *See Appendix 1.*

Using these clusters we used KNN to associate every pass to one cluster and be used in the Xg model. The objective flag of the model was whether the shot ended up as a goal or not (1 or 0). To train the XgBoost model we used the python xgboost library. The parameters were chosen to reduce the risk of overfitting (Max depth : 4, learning rate: 0.05, subsample: 0.9, colsample_bytree: 0.9). The model was programmed to stop after 500 training rounds or 30 rounds without improvement (Chen & Guestrin, 2016). The key metrics for our XgBoost model are: Correlation: 0.2533, AUC: 0.7405 and LogLoss: 0.2197 (Skytte, 2025). Using the shap python library we built a SHAP summary graph to analyze the interaction between features (Lundberg, 2018). *See Appendix 2.*

Next, we needed to compute a pass risk model. To do this we used a very similar setup to the **Xg** model. We also used an XgBoost algorithm and the same parameters to avoid overfitting. To allow more training we made the model stop after 750 steps and not 500 like the Xg model, because after several tries the model would rarely have an early stoppage (Chen & Guestrin, 2016). The features for this model are: the angle in radiants, the nearest defender to the passer, the nearest defender to the intended receiver, the number of defenders in the passing lane, and the x,y coordinates. The 'end' of the pass was simplified by either having a receiver

that successfully or not received a pass as the end coordinates, having a "failedpasslocation" event as the end or a block at the end. If none of these were available we estimated that the pass was directed to the location of the next loose puck recovery. The "nearest defender to" features were calculated using the start and end points of the pass and smallest distance to a defender. The passing lane was calculated as a line that connects the start coordinates to the end and identifies how many defenders are within 2 feet of this lane. These are the key metrics of the model: Correlation: 0.5162, AUC: 0.7989, LogLoss: 0.4478. To see the interaction between features a SHAP graph was built (Lundberg, 2018). *See Appendix 3.*

Based on the **PAV**, we built the passing network metrics. For this part of the analysis, we created several networks working similarly. At their core they all have players as nodes and passes as edges (links between the nodes). The two types of networks are season-long team wide networks and season-long stint wide networks.

For the first network metric we wanted to find, **Out-Degree** and **In-Degree** centralities, we used the team-wide networks. To calculate this we count the number of passes given (**Out-Degre**e) and the number of passes received (**In-Degree**) per player in the offensive zone. We can then add them up and find their total involvement. To normalize the data we also divided the degrees by the time on ice per player times 60 to have In and Out degrees per 60 minutes, the time of a full game. And to further push the data we added PAV of each pass as the weights and normalized it by 60 minutes giving us **InPavPer60** and **OutPavPer60**. The top 5 players for total involvement during the season are in *appendix 4*.

The next metric we wanted to find is **betweenness** centrality. To find this we first created a 'real_stint_id' which represents a specific combination of players from each team. The betweenness centrality was calculated by creating a network in each of these distinct stints. This evaluates the capacity of a player to be the main pass link within a specific stint. We only kept stints with 4 or more players and with 5 or more passes, to have well structured networks (For examples see *Appendix 15-16*). **Betweenness** calculates how often a player falls on the shortest path between nodes. We then weighed all of these results to each player by the amount of times this stint has been together:

$$wBetweenness_j = (\Sigma(betweenness_{ij} * (nStint_i))/(\Sigma\, Stint_j)$$
$$where\ j = playerID\ and\ i = 'real_stint_id'$$

To calculate **betweenness** centrality we used the networkx python library (Networkx, 2014). Inside the network the passes were weighed by PAV. Top 5 weighted **betweenness** in *Appendix 5.*

To calculate **eigenvector** centrality, we took a season-long network instead of a distinct stint. We made a network for each of the teams to figure out if the player was a key driver of the offensive pass network of his team (*see Appendix 9*). We used the networkx library to calculate the **eigenvector** centrality (Networkx, 2014). The weight we used for the edges is: number of passes between two players * (sum of their **pav** - minimum of all **pav**). We multiply by the number of passes to reward players who allow their teammates to pass a lot and we shift the pav scores by the smallest pav to take away all negative values that could break the convergence of the network (Wig et al., 2019). We set the maximum iterations to 200, so that it does not take too long to run and that all networks converge.The top 5 **eigenvectors** in the league can be found in *Appendix 6.*

Next we calculated the **PageRank** (**PR**) score for each of the players. The **PR** score was weighed by **PAV**. Because **PR** cannot handle negative values but the **PAV** can contain negative values we created 2 metrics: the positive **PR** and negative **PR**. The positive **PR** was

calculated using all of the positive **PAV** edges as weights and the negative **PR** was calculated using the negative **PAV** edges. We then computed the **PR** of each using networkx's pagerank function. Then to combine both values we take **pr_net** which is the zscore(**pr_pos - pr_neg**) using scipy.stats zscore function. This gives a normalized score for all players (Godia, 2024). See the top 5 **IN-PAV-per60** in *Appendix 7.*

Now for the final formula:

$$OPIS = w_1 * InPassRisk(InPavPer60) + w_2 * Betweenness + w_4 * Eigenvector + w_5 * PageRank$$

The weights of the final formula can be fine tuned for personal preference of the style of play and importance of each aspect of play. Given the relative range of each metric, we chose the weights: w1 = 1, w2 = 40, w3 = 20, w5 = 5. This gives a relatively even importance across every value and gives us the all around score for importance of the offensive off-puck positioning of the players with a slight importance to PageRank. *Appendix 8* presents the top 5 **OPIS** consult *Appendix 8* while *Appendix 9* shows the bottom 5. To see the importance of every metric given our weights consult *Appendix 10.* To see team histograms examples go to *Appendix 10- to 12.*

Discussion

The first step we took in turning this tracking data into a quantifiable metric of player performance was to recreate an expected goals model, which then served as the foundation for estimating a **Possession Added Value** for passes. Because expected goal modeling has been widely implemented across multiple sports and in hockey as far back as 2005, it provided a reliable baseline upon which we could build our own framework. (Krzywicki, 2005) Rather than relying solely on shot-based statistics, this approach could allow us to evaluate how a passing decision shifts offensive potential with a possession. (McHale, 2018)

However, the **PAV** formulation used in this study relies on a simplifying assumption that the primary alternative to making a pass is taking an immediate shot. To maintain consistency with this assumption, the dataset was pre-filtered to include only passes originating in the offensive zone. As can be seen in Appendix 19., this filtering step improves the interpretability of the **PAV** estimates, as passes from distant zones would otherwise receive systematically large negative values. This simplification naturally biases the model toward rewarding passes that directly increase shot quality. While this aligns with our objective of quantifying offensive threat creation, it may undervalue possession-maintaining or tempo-controlling passes that contribute indirectly to future scoring opportunities. Nonetheless, because quantifying full possession dynamics was not the primary focus of this project, this simplified approach provides a consistent and interpretable estimate of how individual passes shift immediate scoring probability.

We then attacked the passing network by separating the data into analysable sequences of plays, where players are represented as nodes and passes as directed edges linking them. Traditional passing networks reward frequency of involvement but do not distinguish between low-risk perimeter circulation and passes that meaningfully increase scoring probability. (McHale, 2018) The objective of incorporating the pass risk model and **PAV** weighting into the network was therefore to differentiate high-volume passers from high-impact creators. The resulting centrality metrics emphasize players who generate offensive threat rather than merely accumulate touches. (Tyrelstokes, 2021) (Tyrelstokes, 2022)

As shown in *Appendices 4-7*, the top 5 of each metric was computed to assess their validity at highlighting different qualities in players. **Total involvement** highlights players that pass at a high volume but not necessarily high quality passes. **Betweenness** brings out the

players that are the most efficient value distributors. We analyzed this metric from a stint-network perspective to have the players who are efficient within each stint. Then we weighed this by the relative proportion of this stint appearance in the total appearance of the player. This seems to include more defensemen than the other metrics making it a potential impactful weight to tune. **Eigenvector** centrality measures the general influence of a player in its team's passing network. We chose to analyze the eigenvector for the team wide networks for two reasons. The first reason is to make the **eigenvector** algorithm converge and the second was to evaluate the team-wide importance of a player not just within a line. The top 5 in **IN-PAV per 60** (*Appendix 18*) are the players who create more value when receiving passes on a normalized 60 minutes basis. This tends to be forwards since they're the ones usually receiving the puck in higher danger areas (see *Appendix 19*).

The top five players according to the **OPIS** score were all offensive players and appear typically more involved in puck retrieval, transition play, and pass redistribution, which suggests that the metric can capture possession-driving and play-facilitation roles, rather than purely reflecting finishing ability. However, the absence of defenders among the top-ranked players highlights a potential limitation. Because **PAV** is tied closely to immediate scoring impact, defensemen are undervalued since they rarely have high **IN-PAV** due to receiving the puck higher in the offensive zone giving their reception smaller expected goal. This highlights the indirect impacts of the simplification embedded in the **PAV** definition, which potentially prioritizes immediate shot quality shifts over longer possession chains. When we compare two players that had similar seasons (*Appendix 17*), we can see that one person has a significantly higher **OPIS** given similar stats. With an analysis of the **OPIS**, we can conclude that Josh Currie has a better offensive off-puck positioning than Zack Ostapchuk and that he contributes more to the offensive pass network of his line and his team. Now looking at the team-wide graphs (*Appendix 12- 14*), we can conclude that the values are typically better in top ranked teams ('HER') rather than low ranked teams ('LAV' and 'BRI'). We infer that higher ranked teams have more structured play and drive value through more players giving more depth to their **OPIS** ranking.

This type of centrality-based passing network analysis offers several practical applications, as **OPIS** appears capable of identifying structurally important players across the league. A formula such as **OPIS**, or a fine tuned variation of it, could provide coaches and analysts with an additional decision tool, either to confirm the tactical importance of players within a system or to uncover consistently impactful contributors whose influence may not be fully told by normal statistics. Additionally, because the **OPIS** equation has subjective weights, its weights could be adjusted depending on playstyle preferences or optimized through data-driven approaches. Future work could integrate fully functioning **PAV** with simulated passes, or analyze inter-season progression or analysis of playstyle per player. Another area that could be improved is making the analysis rink wide and not just an offensive zone to also capture defensive value. (López Peña, 2012)

Conclusion

This project developed a statistic, **OPIS**, that is passed off of several **PAV**-weighted networks and their corresponding centrality metrics. The objective of this metric is to quantify the offensive off-puck positioning of hockey players. Players with high **OPIS** tend to be the ones creating offensive value flows for their team by occupying being in the right place at the right time. This could help coaches and general managers capture aspects of offensive impact that are not necessarily captured by common statistics. Overall, the proposed formula provides a solid foundation for future work incorporating more advanced underlying metrics. To consult our code see *Appendix 20*.

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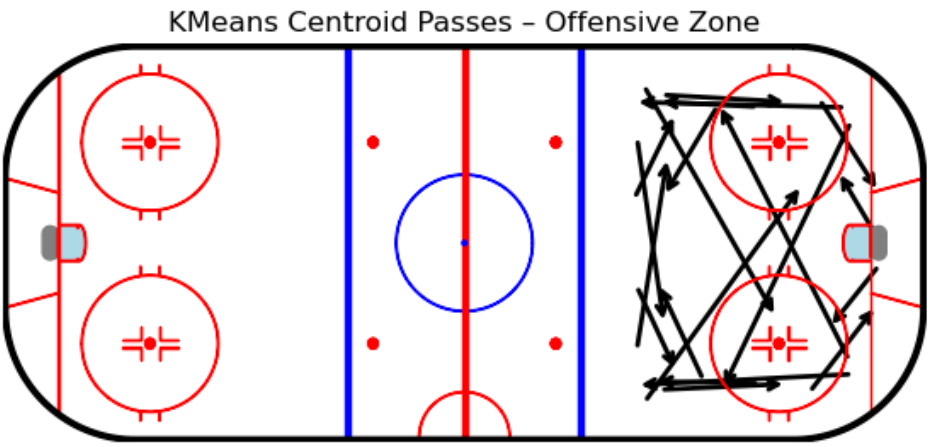
Appendix

Appendix 1:



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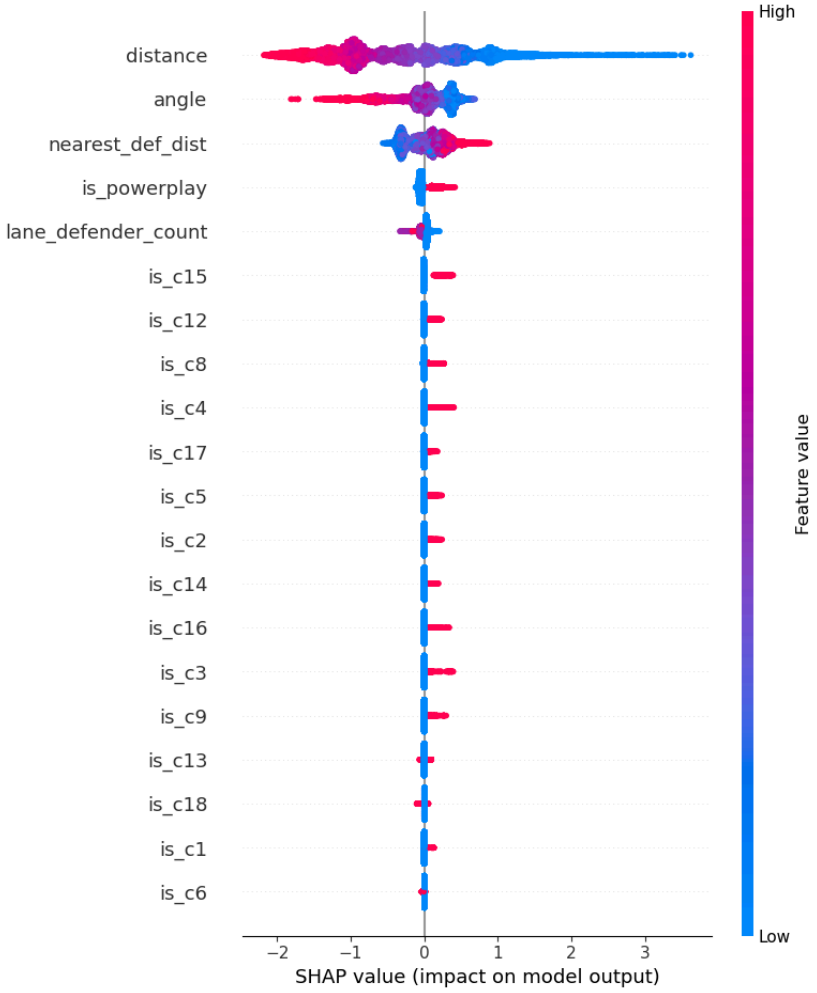

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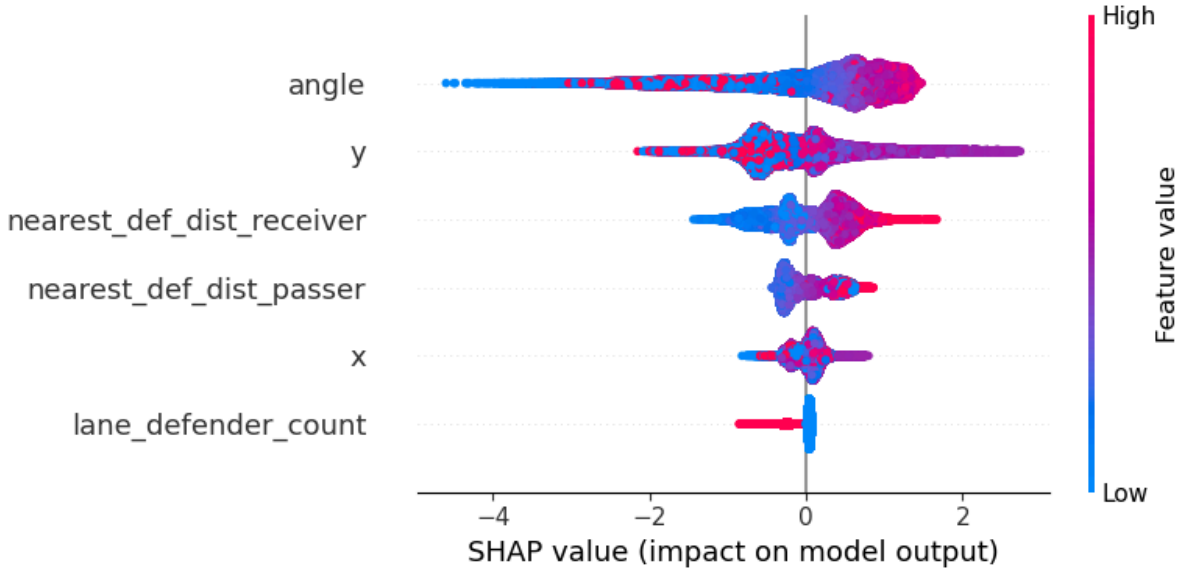


Appendix 4:

Top 5 total_involvement

player_name	out_degree	in_degree	total_involvement
Grimaldi, Rocco	526.0	561.0	1087.0
Smith, Ty	491.0	575.0	1066.0
Tynan, TJ	533.0	434.0	967.0
Marody, Cooper	506.0	445.0	951.0
Goncalves, Gage	445.0	468.0	913.0

Appendix 5:

Top 5 Betweenness

player_name	wBetweeness
Clarke, Brandt	0.6056
Smith, Ty	0.5648
Poirier, Jeremie	0.5256
Larsson, Jacob	0.5237
Wolanin, Christian	0.5154

Appendix 6:

Top 5 Eigenvector Centralities

player_name	Eigenvector
Bourque, Mavrik	0.6585
Griffith, Seth	0.5381
Rempal, Sheldon	0.5218
Malott, Jeff	0.5048
Pederson, Lane	0.5028

Appendix 7:

Top 5 pr_net

player_name	pr_pos_z	pr_neg_z	pr_net
Fogarty, Steven	3.8499	-0.5809	2.3453
Turcotte, Alex	3.4579	-0.5865	2.1551
Thomas, Akil	3.7208	-0.3878	2.1364
Allison, Wade	3.3497	-0.5984	2.1103
Karlsson, Linus	3.3792	-0.545	2.0849

Appendix 8:

Top 5 OPIS

player_name	lnPassRisk	lnPavPer60	Betweenness	eig_pav_team	PageRank	Off_Puck_Impact_Score
Rempal, Sheldon	0.7821	0.9336	0.3461	0.5218	1.3885	31.9522
Bourque, Mavrik	0.7761	1.149	0.2362	0.6585	1.4368	30.6939
Asplund, Rasmus	0.7465	0.8888	0.3288	0.4336	1.4448	29.711
Gawdin, Glenn	0.7965	0.6471	0.2416	0.4835	1.9593	29.6463
Chibrikov, Nikita	0.7983	0.6906	0.4028	0.43	0.7276	28.903

Appendix 9:

Bottom 5 OPIS

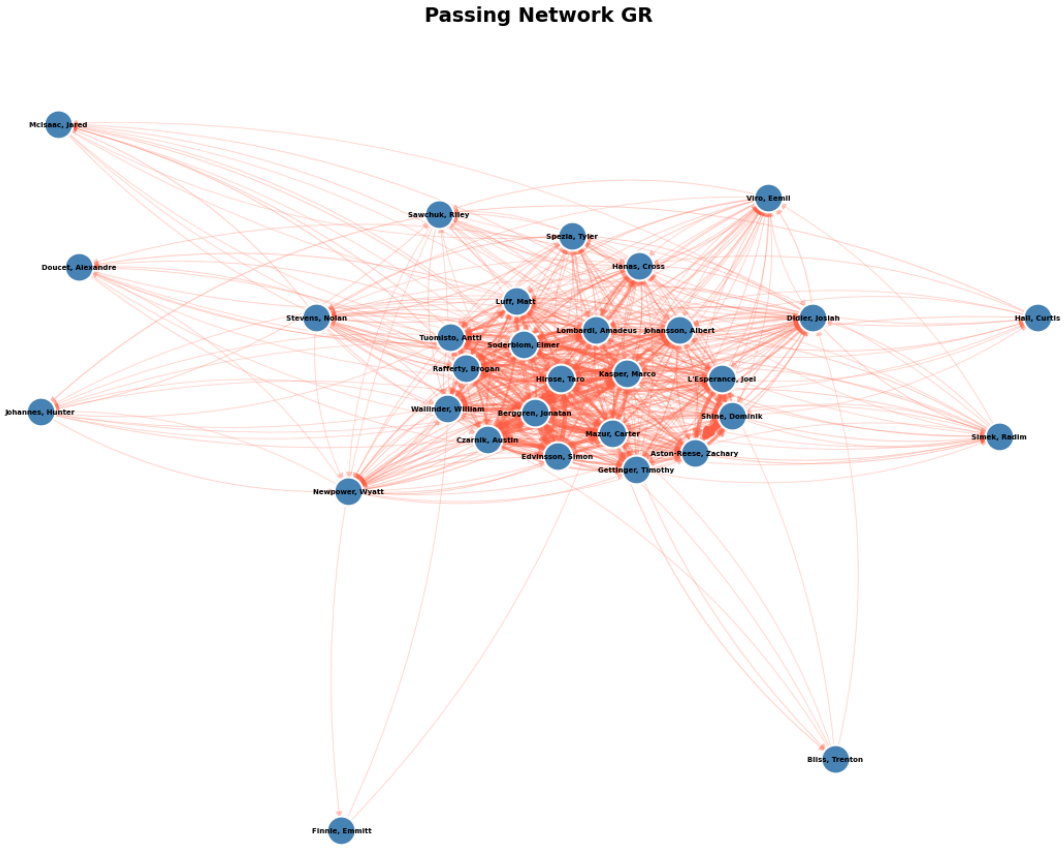
player_name	lnPassRisk	lnPavPer60	Betweenness	eig_pav_team	PageRank	Off_Puck_Impact_Score
Myers, Philippe	0.8712	-0.1625	0.2693	0.001	-6.9052	-23.8773
Sjalin, Calle	0.8724	-0.085	0.2395	0.0239	-5.9338	-19.6866
Smith, Ty	0.8935	-0.6576	0.5648	0.0004	-6.8928	-12.4498
Schuldt, Jimmy	0.8913	-0.1121	0.2027	0.1193	-4.3858	-11.536
Felix, Colin	0.9111	-0.1922	0.0759	0.0376	-2.7206	-9.9893

Appendix 10:

Correllation with OPIS

Metric	Correlation
Off_Puck_Impact_Score	1.0
IN_p%_avg	-0.3399
IN_pav_per_60	0.5202
wavg_bet_stint_1	0.3403
eig_pav_team	0.5314
pr_net	0.7089

Appendix 11:



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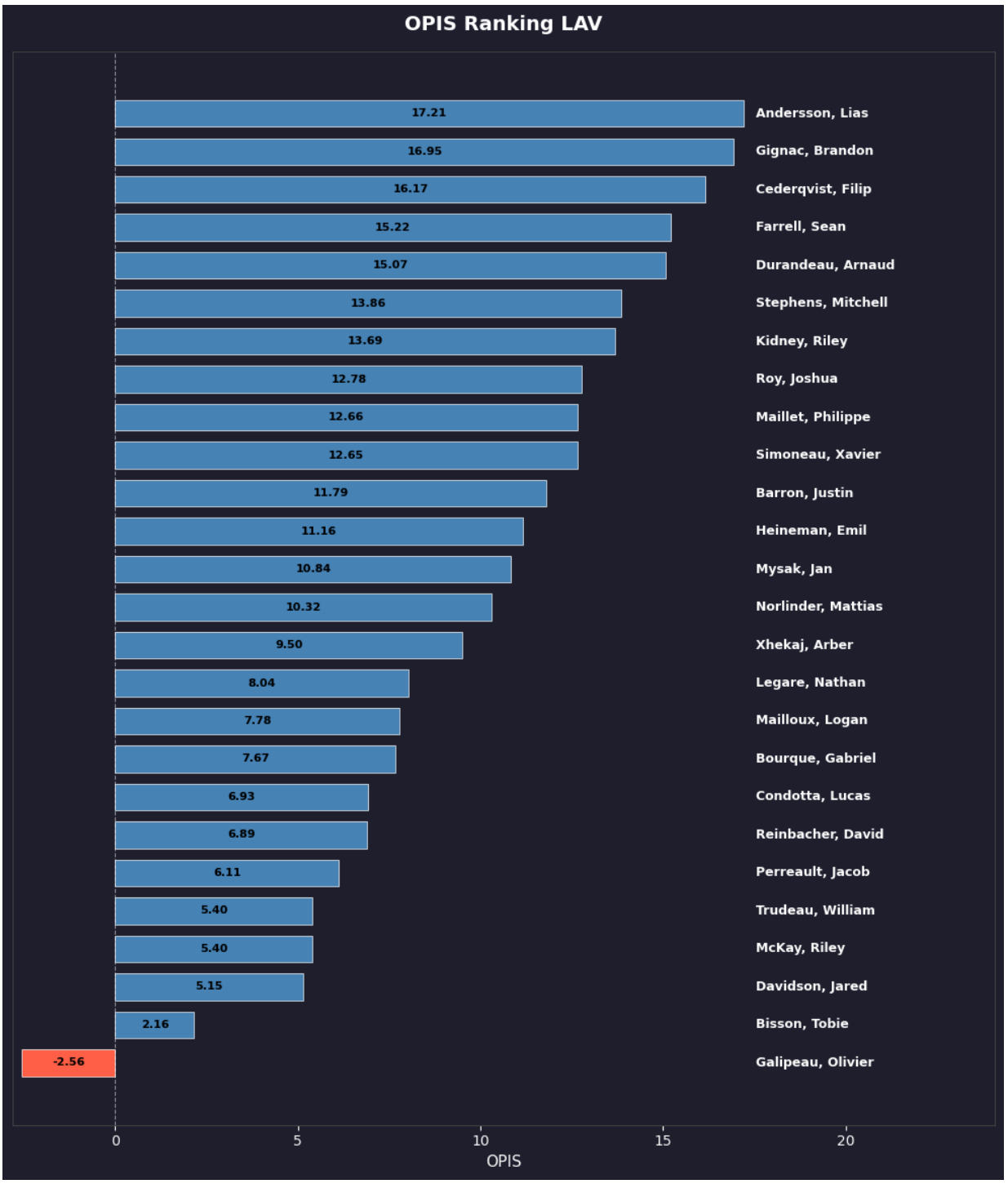
OPIS Ranking LAV
17.21
Andersson, Lias
16.95
Gignac, Brandon
16.17
Cederqvist, Filip
15.22
Farrell, Sean
15.07
Durandeau, Arnaud
13.86
Stephens, Mitchell
13.69
Kidney, Riley
12.78
Roy, Joshua
12.66
Maillet, Philippe
12.65
Simoneau, Xavier
11.79
Barron, Justin
11.16
Heineman, Emil
10.84
Mysak, Jan
10.32
Norlinder, Mattias
9.50
Xhekaj, Arber
8.04
Legare, Nathan
7.78
Mailloux, Logan
7.67
Bourque, Gabriel
6.93
Condotta, Lucas
6.89
Reinbacher, David
6.11
Perreault, Jacob
5.40
Trudeau, William
5.40
McKay, Riley
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Davidson, Jared
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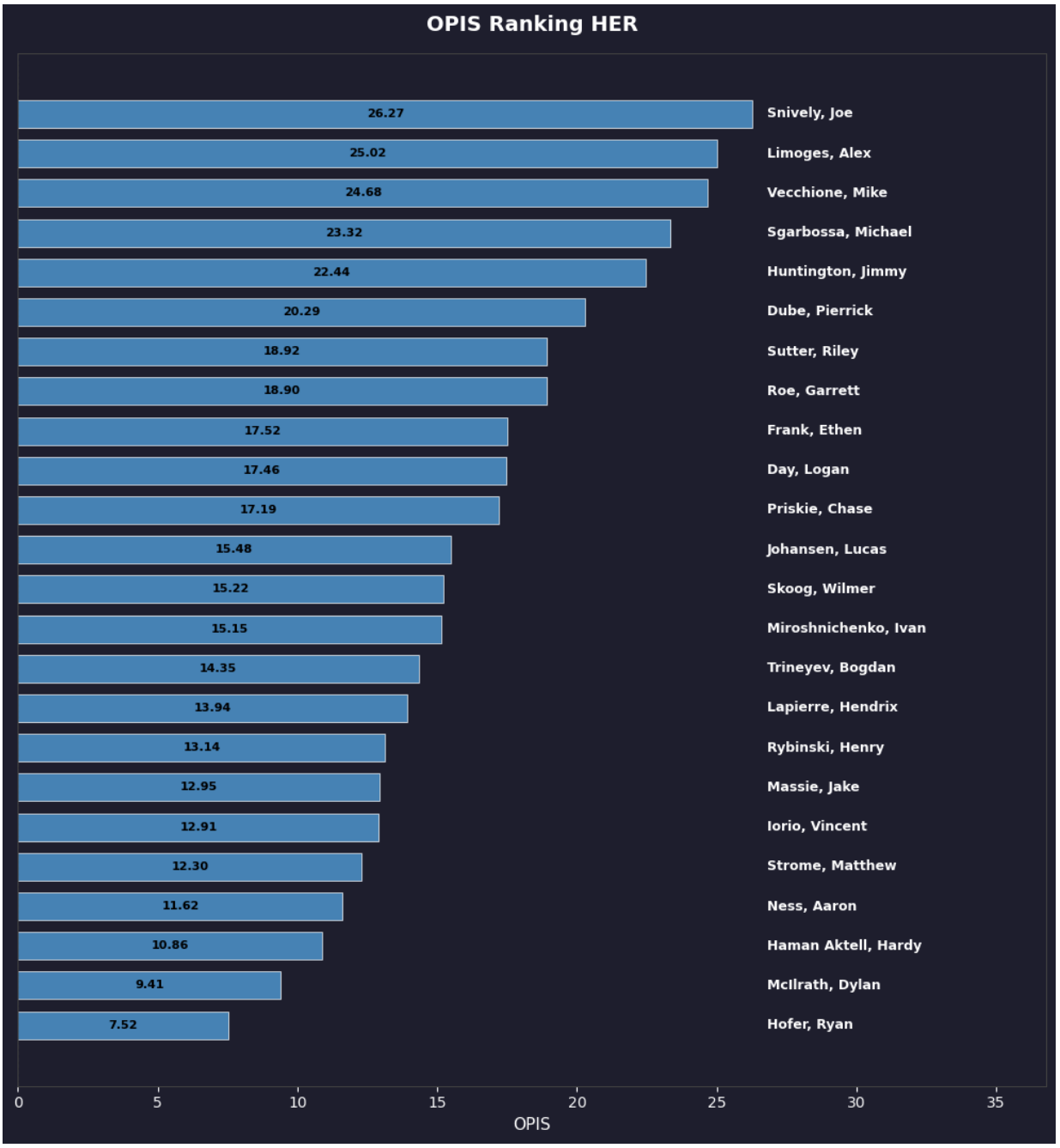
Bisson, Tobie
-2.56
Galipeau, Olivier
0
5
10
15
20
OPIS

Appendix 13:


OPIS Ranking HER
26.27
Snively, Joe
25.02
Limoges, Alex
24.68
Vecchione, Mike
23.32
Sgarbossa, Michael
22.44
Huntington, Jimmy
20.29
Dube, Pierrick
18.92
Sutter, Riley
18.90
Roe, Garrett
17.52
Frank, Ethen
17.46
Day, Logan
17.19
Priskie, Chase
15.48
Johansen, Lucas
15.22
Skoog, Wilmer
15.15
Miroshnichenko, Ivan
14.35
Trineyev, Bogdan
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Lapierre, Hendrix
13.14
Rybinski, Henry
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Massie, Jake
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Iorio, Vincent
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5
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15
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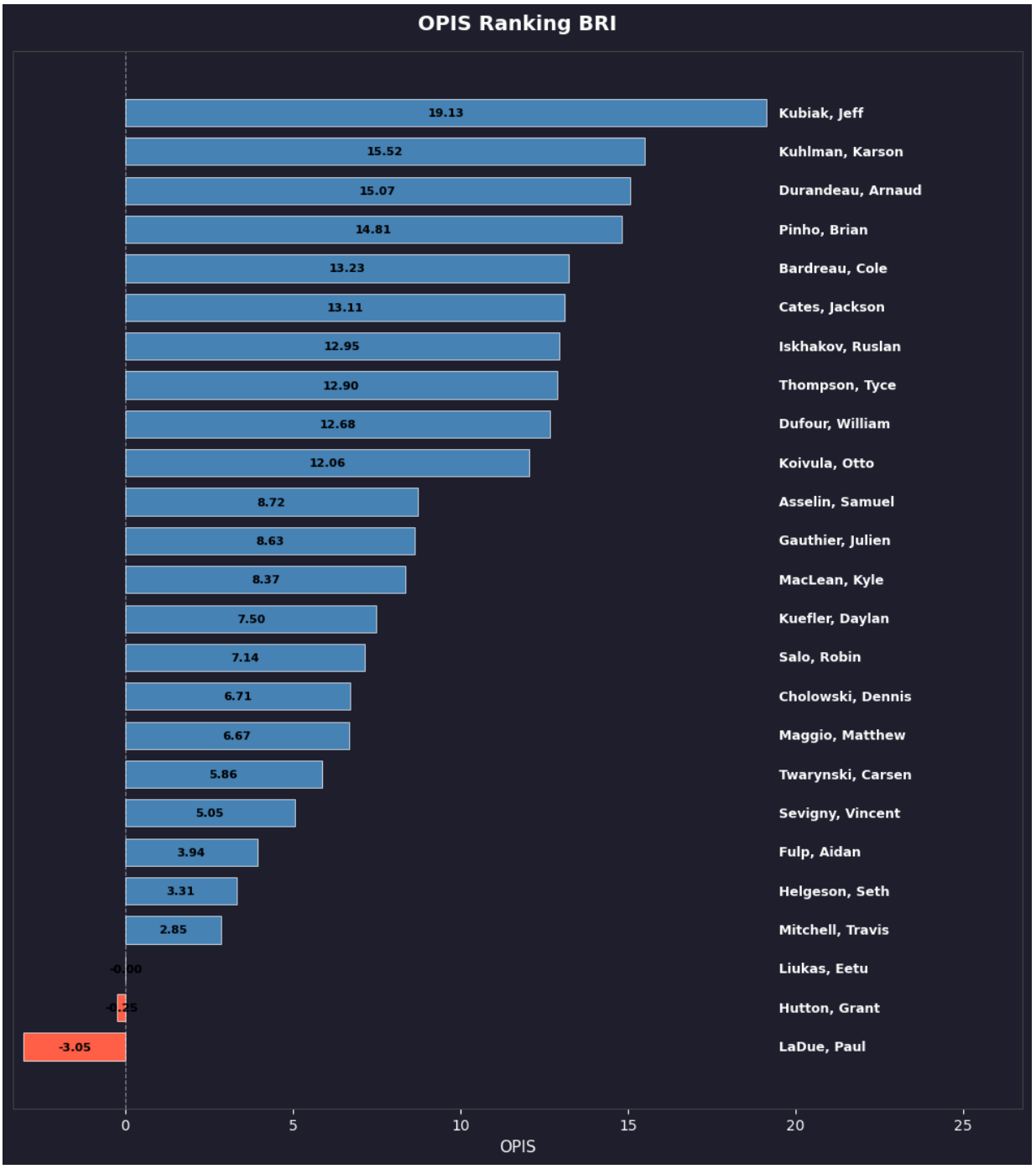
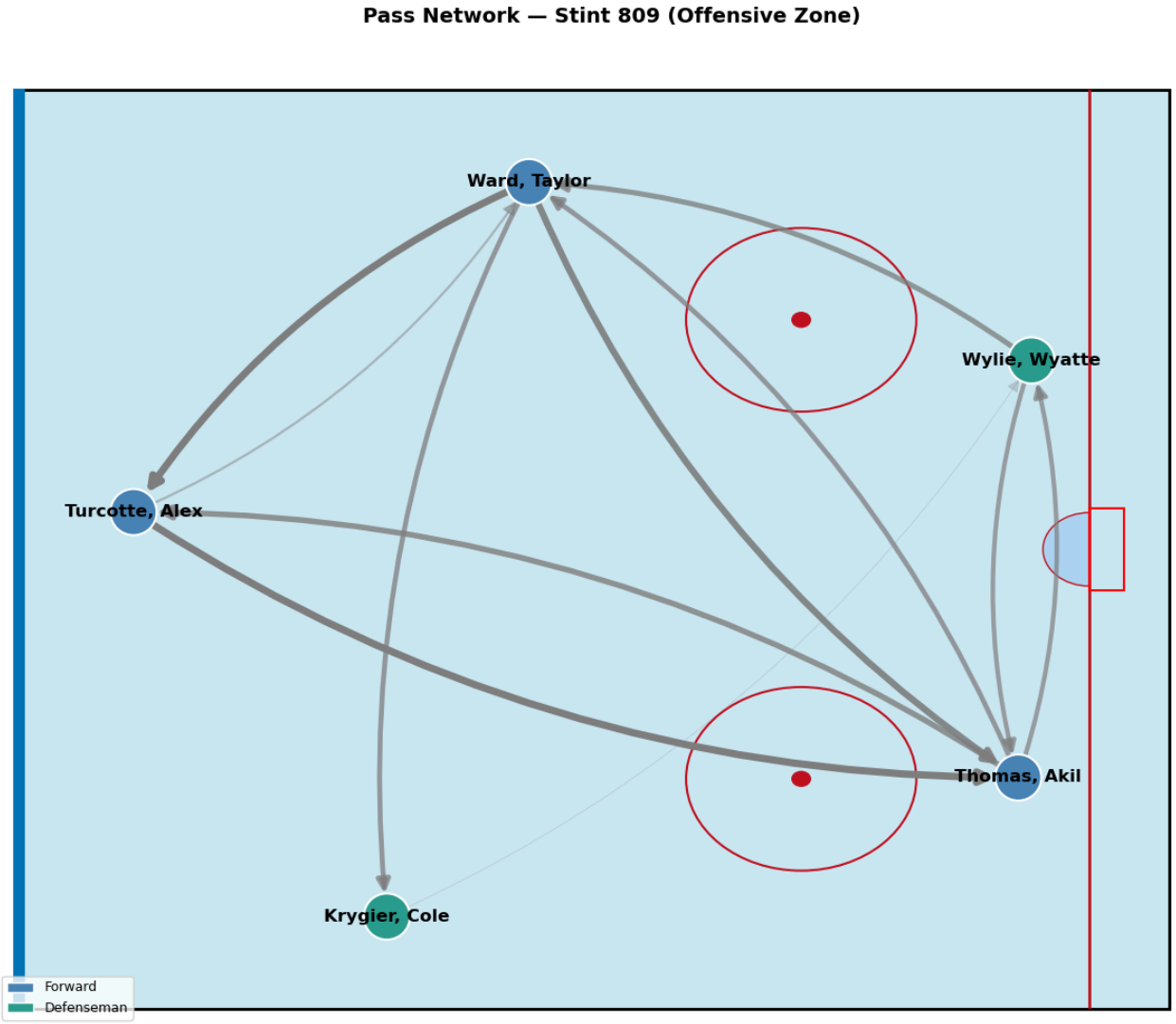
25
30
35
OPIS

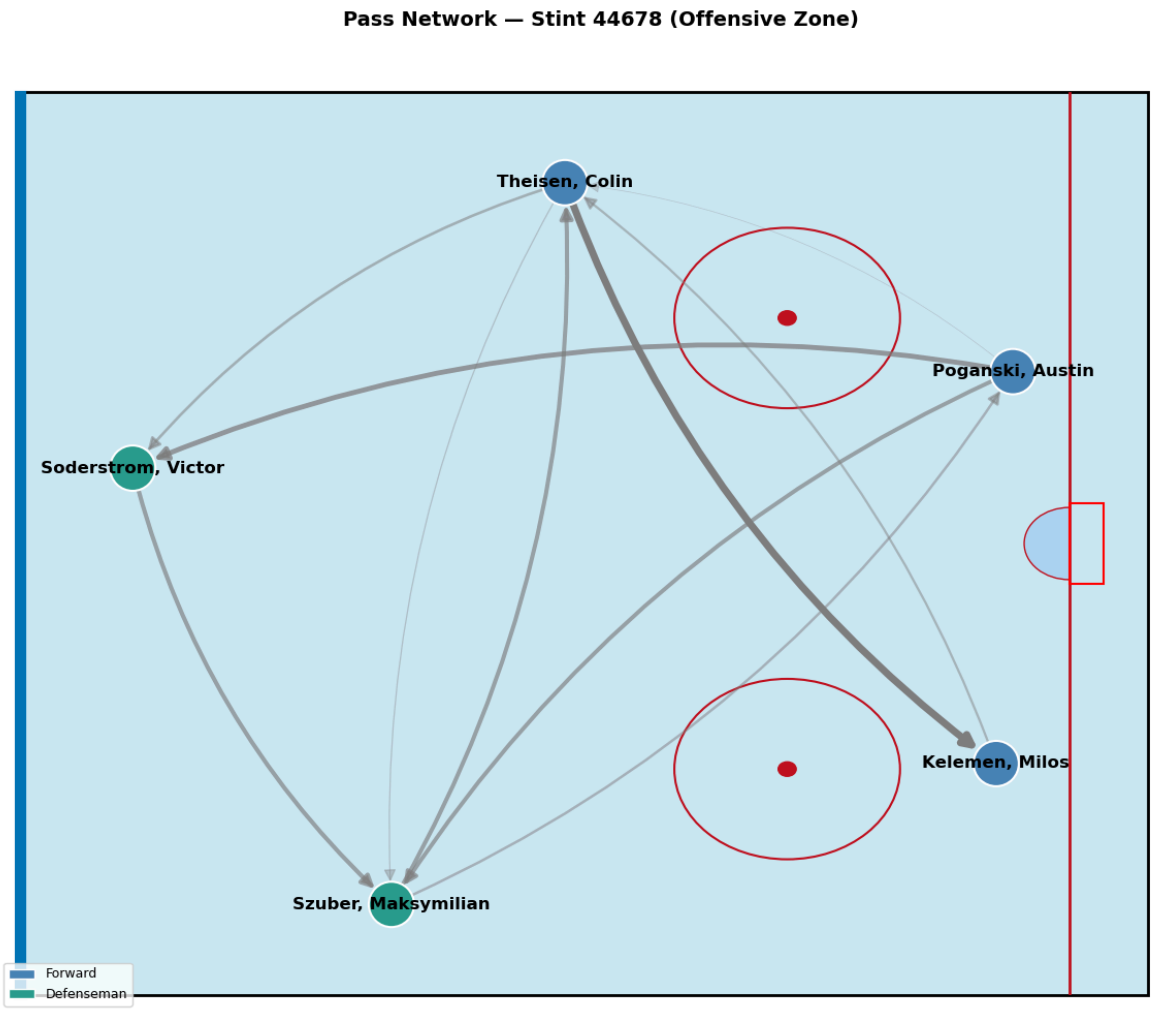
Appendix 14:


OPIS Ranking BRI
19.13
Kubiak, Jeff
15.52
Kuhlman, Karson
15.07
Durandeau, Arnaud
14.81
Pinho, Brian
13.23
Bardreau, Cole
13.11
Cates, Jackson
12.95
Iskhakov, Ruslan
12.90
Thompson, Tyce
12.68
Dufour, William
12.06
Koivula, Otto
8.72
Asselin, Samuel
8.63
Gauthier, Julien
8.37
MacLean, Kyle
7.50
Kuefler, Daylan
7.14
Salo, Robin
6.71
Cholowski, Dennis
6.67
Maggio, Matthew
5.86
Twarynski, Carsen
5.05
Sevigny, Vincent
3.94
Fulp, Aidan
3.31
Helgeson, Seth
2.85
Mitchell, Travis
-0.00
Liukas, Eetu
-0.25
Hutton, Grant
-3.05
LaDue, Paul
0
5
10
15
20
25
OPIS

Appendix 15:


Pass Network — Stint 809 (Offensive Zone)
Ward, Taylor
Wylie, Wyatte
Turcotte, Alex
Thomas, Akil
Krygier, Cole
Forward
Defenseman

Appendix 16:



Appendix 17:

Player Comparison

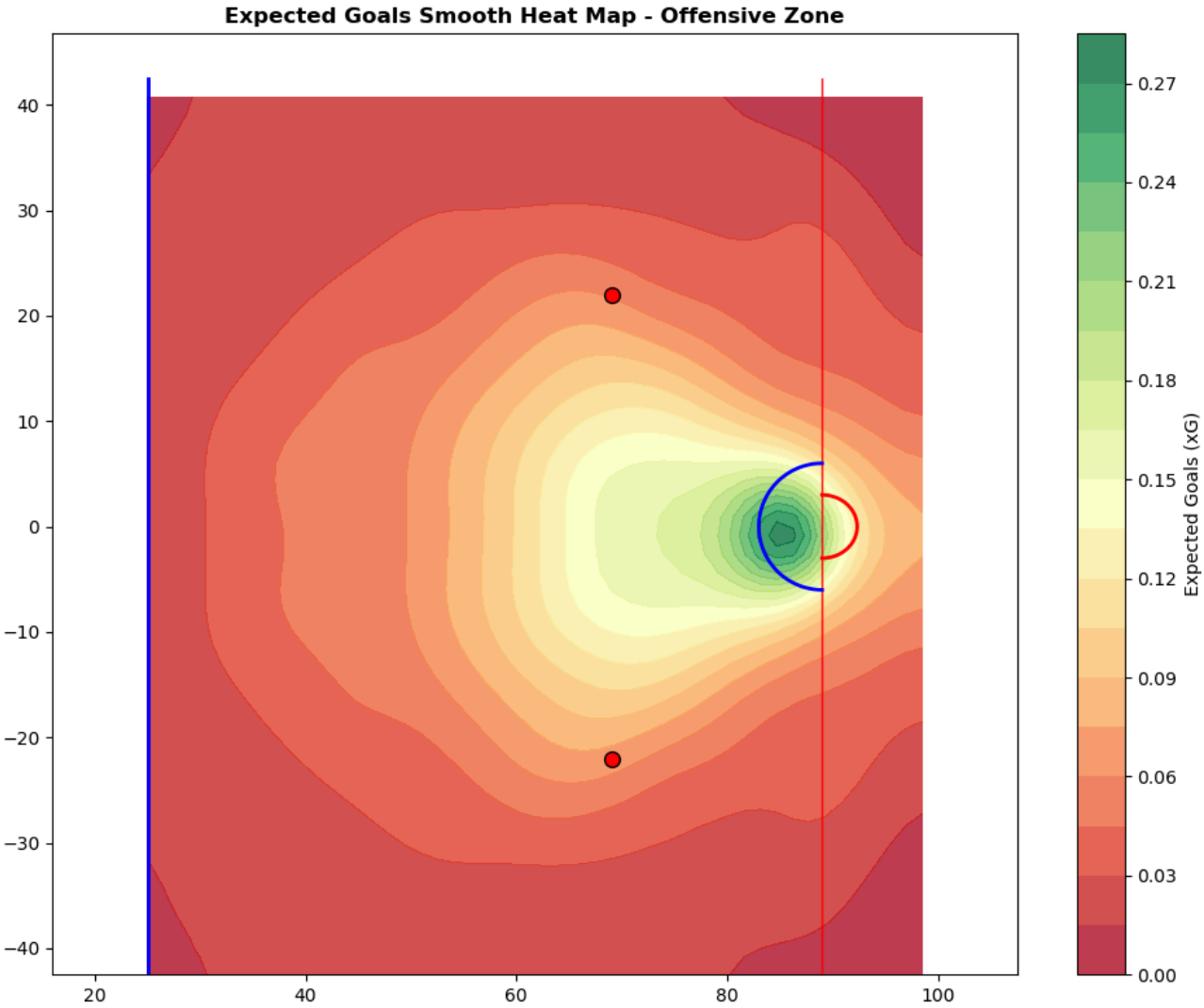
player_name	Off_Puck_Impact_Score	GP	goals	Assist	Points	+/-
Currie, Josh	19.0194	62.0	12.0	17.0	29.0	-1.0
Ostapchuk, Zack	9.4993	69.0	17.0	11.0	28.0	4.0

Appendix 18:

Top 5 IN_pav_per_60

player_name	IN_pav_per_60
Lycksell, Olle	1.5619
Karlsson, Linus	1.5405
Limoges, Alex	1.434
Douglas, Curtis	1.3992
Brazeau, Justin	1.2995

Appendix 19:



Appendix 20:
https://github.com/gabrielchabbert/HALO-Submission